

Automated Visual Triage System for Skin Conditions

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Abstract - Dermatological conditions require timely assessment, but clinical workflows are often bottlenecked by manual visual triage inefficiencies. This project introduces a computer vision application designed to automate the preprocessing, segmentation, and feature extraction of skin lesions from standard digital photographs. Built entirely on foundational image processing algorithms, the system translates raw visual data into standardized representations suitable for medical analysis. The methodology leverages spatial domain intensity transformations to correct inconsistent clinical lighting and morphological filters to suppress noise. By isolating the affected tissue using color-based clustering and extracting robust texture signatures, the application provides an objective baseline for symptom evaluation. This solution ultimately streamlines the diagnostic pipeline, enabling faster and more reliable downstream analysis for healthcare providers.

Index Terms - Automated triage, Computer vision, Image segmentation, Skin lesions

I. INTRODUCTION

The initial assessment of dermatological issues relies heavily on subjective visual inspection, which is frequently compromised by poor image quality, inconsistent illumination, and human cognitive bias. To address this bottleneck in healthcare delivery, a reliable, automated tool is required to standardize patient images and mathematically quantify lesion characteristics. This project addresses these inconsistencies by implementing a sequential computer vision pipeline.

II. METHODOLOGY

A. I. Preprocessing and Enhancement

Clinical images often suffer from uneven lighting. To correct this, the system normalizes clinical photo brightness and contrast utilizing spatial domain techniques such as histogram equalization and contrast stretching.

B. II. Noise Reduction and Edge Detection

High-frequency image artifacts are suppressed through the application of a Gaussian blur filter. Once the noise is mitigated, the Canny edge detection algorithm is applied

to identify and outline the structural boundaries of skin anomalies.

C. III. Isolation and Segmentation

To ensure the analytical phase only processes relevant data, K-means clustering is employed. This color-based segmentation partitions the image, effectively isolating the lesion from the surrounding healthy tissue.

D. IV. Feature Analysis

Finally, the system generates a quantitative texture profile of the segmented region using the Local Binary Pattern (LBP) descriptor. This robust feature extraction method provides a numerical baseline that can be easily parsed by diagnostic algorithms or human reviewers.

III. CONCLUSION

By translating raw visual data into a standardized, noise-free representation, this automated triage pipeline bridges the gap between primary image capture and advanced medical analysis, demonstrating a highly practical application of foundational computer vision techniques.

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